CP1.7: Linear Transformations

1. Representing Transformations as Matrices

A **linear transformation** in 2D maps every point $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$ to $\mathbf{x}' = M\mathbf{x}$ where M is a 2×2 matrix.

Key principle: The columns of the transformation matrix M are the images of the standard basis vectors:

$$M = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{pmatrix}$$

where $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

To find the matrix of any transformation: find where $(1, 0)$ maps to — that gives column 1. Find where $(0, 1)$ maps to — that gives column 2.

2. Standard Transformation Matrices

Transformation	Matrix	Notes
Reflection in x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$x \to x,\ y \to -y$
Reflection in y-axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	$x \to -x,\ y \to y$
Reflection in $y = x$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	Swap x and y

© RAG Learning · Not for redistribution

Reflection in $y = -x$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	Swap and negate
Rotation anticlockwise by θ	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$	About the origin
Rotation by 90° anticlockwise	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	Special case $\theta = 90^\circ$
Rotation by 180°	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	Half turn
Rotation by 270° anticlockwise	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$	Same as 90° clockwise
Enlargement, scale factor k	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$	Centre at origin
Stretch by factor a in x, b in y	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	Axis-aligned stretch
Shear in x-direction, factor k	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$	y unchanged; $x \to x + ky$
Shear in y-direction, factor k	$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$	x unchanged; $y \to kx + y$

Example: Find the matrix for reflection in the line $y = x\tan\theta$ (a line through the origin at angle θ to the x-axis).

The general reflection matrix in the line at angle θ is:

$$M = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$$

Check: for $\theta = 45^\circ$ (line $y = x$): $\cos 90^\circ = 0$, $\sin 90^\circ = 1$, giving $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ✓

3. Unit Square Under Transformation

A useful way to visualise a transformation is to track the unit square with vertices at $O = (0, 0)$, $A = (1, 0)$, $B = (1, 1)$, $C = (0, 1)$.

Notice that for the shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$: the area of the parallelogram equals the area of the original square (determinant = 1). This illustrates the geometric meaning of the determinant.

4. Combining Transformations

To apply transformation T_1 *followed by* transformation T_2, the combined matrix is $M_2 M_1$ (multiply in **reverse order**).

If T_1 has matrix M_1 and T_2 has matrix M_2, then the image of point $\mathbf{x}$ under both transformations is:

$$\mathbf{x}' = M_2(M_1\mathbf{x}) = (M_2 M_1)\mathbf{x}$$

Exam tip: The order of matrix multiplication is critical. "Apply T_1 then T_2" means the combined matrix is $M_2 M_1$, NOT $M_1 M_2$. A very common error. The transformation applied FIRST goes on the RIGHT.

Example: Find the matrix for "reflect in the x-axis, then rotate anticlockwise by 90°".

M_1 (reflect in x-axis): $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

M_2 (rotate 90° anticlockwise): $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Combined: $M_2M_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

This is the reflection in $y = x$!

5. Finding Images of Points and Lines

Image of a Point

To find where point (a, b) maps to under transformation M:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = M\begin{pmatrix} a \\ b \end{pmatrix}$$

Image of a Line

To find the image of a line $y = mx + c$:

1. Let a general point on the line be $\begin{pmatrix} x \\ mx + c \end{pmatrix}$
2. Apply the transformation: $\begin{pmatrix} x' \\ y' \end{pmatrix} = M\begin{pmatrix} x \\ mx + c \end{pmatrix}$
3. Express x and $mx + c$ in terms of x' and y'
4. Eliminate x to find the equation of the image line

Example: Find the image of the line $y = 2x + 1$ under the transformation $M = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$.

A point on the line: $\begin{pmatrix} t \\ 2t + 1 \end{pmatrix}$.

Image: $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} t \\ 2t + 1 \end{pmatrix} = \begin{pmatrix} 2t + 2t + 1 \\ 3(2t + 1) \end{pmatrix} = \begin{pmatrix} 4t + 1 \\ 6t + 3 \end{pmatrix}$

So $x' = 4t + 1$ and $y' = 6t + 3$.

From x': $t = \frac{x'-1}{4}$. Substitute: $y' = 6 \cdot \frac{x'-1}{4} + 3 = \frac{3(x'-1)}{2} + 3 = \frac{3x'-3+6}{2} = \frac{3x'+3}{2}$
Image line: $y = \frac{3}{2}x + \frac{3}{2}$

6. Invariant Points and Lines

An **invariant point** is a point that maps to itself under the transformation:
$M\mathbf{x} = \mathbf{x}$, i.e., $(M - I)\mathbf{x} = \mathbf{0}$

An **invariant line** is a line that maps to itself (as a whole line, not necessarily point by point). Every point on the line maps to another point on the same line.

Finding Invariant Points

Solve $M\mathbf{x} = \mathbf{x}$, which is the same as finding the eigenvalue 1 eigenvectors (if they exist). For most transformations, the only invariant point is the origin.

Finding Invariant Lines

Assume the line has equation $y = mx + c$. Apply the transformation to a general point, and require the image to also lie on $y = mx + c$. Solve for m and c.

Example: Find the invariant lines (of the form $y = mx$) of the transformation $M = \begin{pmatrix} 3 & 1 \\ 4 & 0 \end{pmatrix}$.

On the invariant line $y = mx$, take point $\begin{pmatrix} t \\ mt \end{pmatrix}$.
Image: $M\begin{pmatrix} t \\ mt \end{pmatrix} = \begin{pmatrix} 3t + mt \\ 4t \end{pmatrix} = t\begin{pmatrix} 3+m \\ 4 \end{pmatrix}$

For the image to lie on $y = mx$: $\frac{4}{3+m} = m \Rightarrow 4 = m(3+m) = 3m + m^2$
$m^2 + 3m - 4 = 0 \Rightarrow (m+4)(m-1) = 0 \Rightarrow m = -4$ or $m = 1$
Invariant lines: $y = x$ and $y = -4x$.

7. 3D Transformations

In 3D, a linear transformation maps $\mathbf{x} \in \mathbb{R}^3$ to $M\mathbf{x}$ where M is a 3×3 matrix. The same principles apply:

- Rotation about the x-axis by θ: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$
- Rotation about the y-axis by θ: $\begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$
- Rotation about the z-axis by θ: $\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$
- Reflection in the xy-plane: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

The $|\det(M)|$ gives the volume scale factor for 3D transformations.

8. Summary

Task	Method
Find transformation matrix	Find images of $(1, 0)$ and $(0, 1)$ — these are the columns
Image of point (a, b)	Compute $M \begin{pmatrix} a \\ b \end{pmatrix}$
Image of a line	Parameterise, apply M, eliminate parameter
Combine: T_1 then T_2	Matrix: $M_2 M_1$ (right to left)
Invariant points	Solve $M\mathbf{x} = \mathbf{x}$
Invariant lines $y = mx$	Apply M, require image on same line; solve for m
Area scale factor	$\lvert\det(M)\rvert$